

# Ganesh Gupta

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## EXPERIENCE

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Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Engineered pluggable **distributed executor** routing pipeline nodes to **Docker** or **AWS Lambda**, backed by **S3** artifact store eliminating redundant recomputation.
- Wrote systems-level code in **Go**, **Rust**, and **C++** for **scheduling**, **resource allocation**, and **fault-tolerant placement** under GPU-scarcity constraints.
- Designed stateful recovery via per-node **Redis checkpointing**; interrupted workflows **resume from the last committed step** across **distributed execution**.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Built **model observability** framework tracking **drift**, latency, and **eval regressions** in production, enabling governance-aligned **rollback** workflows.
- Built and maintained **Airflow DAGs** orchestrating multi-stage data pipelines on **GCP**, with **dependency management** and automated **failure alerting**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Built and operated **25+ production microservices** in **C++** and SQL-backed **distributed systems** for **1M+ users**, sustaining **99.9% uptime** over 4 years on **AWS** (EC2, RDS, SQS).
- Standardized failure-handling contracts across **25+ financial services**, reducing production escalations **60%** in one quarter via consistent **retry**, **timeout**, and **propagation** semantics.
- Designed high-throughput SQL pipelines and **distributed messaging** workflows on **AWS SQS**, cutting nightly batch aggregation from **8 minutes to 90 seconds** via indexed views and partitioning.

## PROJECTS

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Distributed KV Store / VPC Control Plane Simulator

[github.com/ganeshhgupta/distributed-kv-store](https://github.com/ganeshhgupta/distributed-kv-store)

- Raft-consensus** KV store with **linearizable reads** and **log compaction**; companion **VPC control plane** simulator modeling subnet routing and security groups.

RootCause AI

[github.com/ganeshhgupta/rootcause-ai](https://github.com/ganeshhgupta/rootcause-ai)

- LangGraph** multi-agent system diagnosing **production incidents** via tool-use chains; routes alerts to specialized sub-agents and synthesizes **root cause reports**, reducing **MTTR**.

LangFetch

[tinyurl.com/LangFetch](https://tinyurl.com/LangFetch)

- LangGraph** agent decomposes multi-table questions into dependency-ordered **SQL subqueries** with intermediate result passing; deployable as **MCP-compatible** server.

## EDUCATION

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University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

## SKILLS

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**Languages:** Python, Go, C++, Rust, Java, SQL

**Infra/DevOps:** Docker, Kubernetes, Terraform, CI/CD, Observability

**Cloud:** AWS (Lambda, S3, SQS, EC2), GCP

**Practices:** Distributed Systems, GPU Fleet Ops, Incident Response

## CERTIFICATIONS

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AWS Certified Developer Associate | GCP Professional MLOps Engineer